



Copper ax
blade tied to a
wooden handle



◀ PRECIOUS METALS

Gold and silver are shiny, rare metals that have been used to make jewelry since ancient times. These metals offered people a way to display wealth and high status. Around 4000 BCE, high-ranking people at Varna in modern-day Bulgaria were buried in graves filled with gold ornaments.

FACT

Ancient Egyptians found small amounts of pure iron in meteorites—rocks from space that hit Earth. They called this “metal of heaven.”

Working with metals

Early humans made use of stone tools, but somewhere between 6000 and 2000 BCE, depending on where they were in the world, people began to use metals widely, bringing an end to the Stone Age. Unlike stone, metals could be shaped into any form by molding, beating, and sharpening. Some metals, such as gold and silver, were also prized for their beauty.

Metalworking developed in stages, as people gradually learned how to work harder metals that proved to be more useful. Early people used naturally occurring nuggets of soft metals, such as lead and copper, to make jewelry. Later, they learned to extract these metals from their ores (metal-bearing rocks) by heating them in a process called smelting.

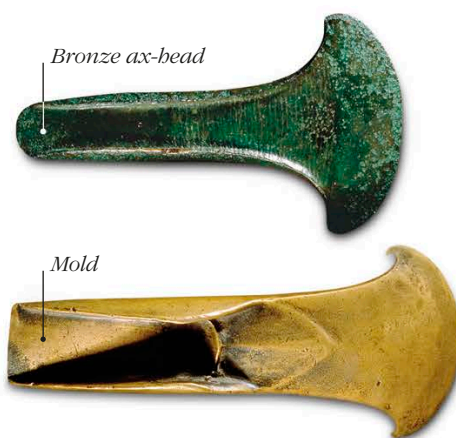
The earliest metal tools were made from copper. In the late 4th millennium BCE, people in western Asia discovered that adding a small amount of tin to copper created a

harder material called bronze. As tin is rare, demand for it gave rise to long-distance trade, creating great wealth for the societies that mined and worked it.

Iron, the most common metal, was the last to be worked. It is even harder than bronze and extracting it required new techniques that were invented in western Asia around 1500 BCE. Iron changed everyday life, and was soon used to make sturdy tools, weapons, pots, and nails, among many other useful items.

COPPER TOOLS ▶

The first metal tools were copper chisels, which were used to build the Egyptian pyramids. Copper is a soft metal, so these tools needed constant resharpening. This well-preserved ax belonged to Ötzi the Iceman, whose frozen, 5,300-year-old body was found in the Alps.



Bronze ax-head

Mold

▲ BRONZE

Bronze was used to make axes by melting copper and tin together and pouring the molten mix into molds. It was hard and long-lasting, which led to the invention of new weapons, such as the sword. Bronze could also be hammered into sheets to make armor.